

Recoverable but unrealized: the mask-quality gap of prototype-based instance segmentation

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Abstract

A correctly localized instance can still fail a strict mask criterion. We ask whether such mask failures are recoverable inside the frozen representation and, if so, why the native model does not recover them. On full COCO val2017 we first account for where mask quality is lost across raw candidate geometry, class-argmax conditioning, and output retention. We then show that 75.0% of strict argmax-conditioned mask failures admit a coefficient correction that reaches Mask75 with the prototype bank frozen, under the project's documented finite-oracle definition re-derived and independently verified in this study (66.0% under the original panel's partially recorded decode conventions; Section 4.2). Seven controlled comparisons exclude the usual explanations: oracle-target supervision does not transfer to held-out candidates through the native coefficient branch, and a constrained functional projection retains less oracle gain than a random support control. The bottleneck is therefore located in how coefficients are realized, not in whether the information exists. All readouts are diagnostics on a single frozen checkpoint; no accuracy claim is made.

Keywords: Instance segmentation ; Error diagnosis ; Prototype coefficients ; Mask quality ; Frozen-model analysis

1. Introduction

Modern prototype-based instance segmenters decode masks from a shared prototype bank and per-instance coefficients (?). A candidate can localize its object well and still fail a strict mask criterion. Standard aggregate accuracy does not say where the quality is lost, and observing excess foreground does not say which upstream stage produced it. We separate the question into layers: whether the raw candidate pool contains the capability at all, whether semantic conditioning keeps it, and what the output stage retains (?).

Our question is sharper. Among instances that are class-argmax correct and Box75 correct, what share of mask failures could be repaired by changing only the coefficient vector inside the frozen prototype space, and if that share is large, why has the native model not realized it? We find that 75.0% of the strict population is recoverable in exactly this sense (66.0% under the original panel's recorded conventions). The first half is a recoverability fact; the second half is an exclusion problem. We answer both, and the answers change where a fix has to act.

The paper's claim is deliberately negative in a useful way: the recoverable share is large, and none of the seven candidate mechanisms we test accounts for the gap. Short-budget supervision toward oracle directions does not transfer to held-out candidates; a constrained functional projection of the same oracle directions protects reliable regions extremely well but retains only about a third of the available gain, and less than a random-support control. No test yields a deployable method; each is a controlled comparison that removes one explanation.

2. Related work

Prototype-coefficient decoding is established (?). Error-analysis tooling for detection and segmentation is established (?). Boundary refinement and quality scoring are established (????). Ownership and competition between instances has been modeled before (?). We therefore claim no structural novelty. What is new here is the combination: a strict argmax-conditioned failure population, a frozen-prototype recoverabil-

ity measurement on the full validation set, and a chain of controlled exclusions that together localize the gap.

3. Accounting: where mask quality is lost

3.1. Four evaluation units

Let a raw candidate be identified by image, branch, pyramid level, grid position and input shape. For each ordinary GT we ask, in order: (i) does any raw position achieve Box75 and Mask75; (ii) does any raw whose predicted-class argmax equals the GT class do so; (iii) does the normal output retain a good candidate. The units answer different questions and are never interchanged. This yields the population used below: instances that are argmax-correct and Box75-correct yet have no argmax-correct Mask75 candidate anywhere.

3.2. Population and baseline

On COCO val2017 (?) with 5,000 images and 36,335 ordinary GT, the strict population contains 6,459 instances (17.78%; A1 gives the score-independent geometry classes, A4 the baseline AP). All readouts use the official frozen YOLO26m-Seg checkpoint (?). Output-stage retention accounts for a further, separately recorded loss of otherwise good masks (A3). Table 1 collects the three layers. The remainder of the paper studies only the strict population.

4. Recoverability inside the frozen prototype space

4.1. Oracle definition

Fix the prototype bank, the coefficient of the candidate, its predicted box and the decoding path. For each instance solve, in the GT-box region of the input grid,

$$\Delta c_{\lambda}^* = \arg \min_{\Delta c} \frac{1}{A} \sum_u \text{BCE}(p_u^T(c_0 + \Delta c), y_u) + \frac{\lambda}{2} \|\Delta c\|_2^2, \quad c^* = c_0 + \Delta c_{\lambda}^*, \quad (1)$$

with λ fixed in advance; Eq. 1 is the project's recorded finite-oracle convention and is GT-assisted by construction. An instance is *recoverable* when some argmax-correct raw's c^* reaches Mask75 after the official decode.

Table 1. Where mask quality is lost, in three separate accounting layers (COCO val2017, 5,000 images, 36,335 ordinary GT, frozen YOLO26m-Seg). The layers answer different questions and are never interchanged. Shares are of 36,335 GT except in the output-retention block, where they are of the 13,568 GT whose normal output fails Mask75.

layer	state	GT	share
raw pool (score-free)	same raw Box75+Mask75	23,805	65.52%
	good box, no good mask in pool	6,688	18.41%
	good mask, no good box	487	1.34%
	good box/mask on different raws	531	1.46%
	neither good	4,824	13.28%
argmax-conditioned	semantic fail	44	0.12%
	box fail	6,335	17.43%
	mask fail (strict)	6,459	17.78%
	box/mask misaligned	484	1.33%
	success	23,013	63.34%
output retention	no raw Mask75 (of 13,568 output failures)	11,512	84.85%
	good mask, class score ≤ 0.001	1,797	13.24%
	not in head top-300	153	1.13%
	per-category maxDets budget	103	0.76%
	COCO matching competition	3	0.02%
baseline	Mask AP / Mask75 matched	43.518 / 22,767 (62.66%)	

4.2. Result

We re-derived the whole panel under Eq. 1 with the letterbox geometry, GT raster and decode path rebuilt from the retained per-image records, and verified the re-derivation by independent re-solves from different initializations, exact re-aggregation, and identity cross-checks. The re-derived share is **4,843 of 6,456 instances (75.0%)**; Table 2 reports strata by GT size next to the original panel’s recorded values. The original panel’s solver is not part of the retained record, and its decode and ROI conventions are only partially documented; we therefore report both values and use neither as the sole basis of any claim. Three GTs (0.05%) whose GT box degenerates on the decode grid are recorded as skipped. The difference concentrates in small instances and is consistent with a threshold-boundary convention effect near the 0.75 criterion, not with a different oracle family.

An independent panel re-measures the same construct. The remainder of the non-recoverable set splits into fixed-prototype/decoding-limited and support-infeasible instances in the original record (2,169 and 27 respectively), and 309 candidate rows of that record store the support ceiling rather than a solved value; these accounting facts do not change either recoverable count. Position selection is not the explanation: across instances with several argmax-correct raws, the oracle best exceeds the per-instance median by only 1.293 points on average (1.854 on the 4,504 instances that have multiple raws). Whatever convention is used, the recoverable share is a large majority of the strict population, and no fitted mechanism below realizes it.

5. Why the native model does not realize it

Seven comparisons each remove one explanation. Table 3 summarizes; we state each with its boundary.

Supervision quantity. Adding a spatial-logit teacher toward oracle logit changes moves the training objective but not the output: the controlled comparison is +0.013 points of image-macro IoU against its same-budget control, with the target group’s interval crossing zero and no additional Mask75 crossings.

Ownership readout and solver. Reading pixel ownership from frozen neck features does not beat a same-capacity control; routing external spatial evidence through the fixed solver stays below the native model.

Shared final layer. Re-fitting the shared affine final layer on frozen features recovers 46.8% of the diagnostic objective gap and, when frozen and applied to normal images, yields a bootstrap-positive real-mask improvement (+0.055 image-macro IoU), concentrated in P4/P5. The remaining per-instance oracle gap is not recoverable by any shared map.

Post-hoc global correction. A frozen global corrector improves image-macro IoU by 0.092 points, CI [+0.022, +0.167], with no reliable Mask75 change.

Oracle-target distillation. Fine-tuning only the coefficient heads toward oracle-derived targets does not transfer: on held-out candidates both the target arm and a same-budget BCE-only control move *against* the oracle direction (cosine -0.03 ; -0.12 at P5 and for large instances), the repair gain is identical with and without the oracle target, and both damage previously correct instances at about 7%.

Constrained functional projection. Projecting oracle corrections onto a trust-region subspace (evidence pixels by current-mask uncertainty, protected pixels by margin) suppresses protected-zone logit change to 6% of the oracle’s own, but retains 30% of the oracle IoU gain against an 80% gate, and retains less than a random same-cardinality support control (59%).

Reading. The gap is not information availability (the recoverable share is large) and not any of the seven tested mechanisms. It sits at the realization stage: the native coefficient head’s function class, as trained, does not select the recoverable directions.

6. What the gap implies for corrections

Two independent bounding results frame what a correction can buy. Removing false-positive pixels from already-correctly-boxed predictions raises mask AP by 10.999 points in a GT-assisted ideal repair, and the failure rate varies strongly with the GT’s fill ratio (62.4% vs 10.5% between the lowest and highest fill strata), which is an object-selection signal, not a mechanism. Decode-level substitutions do not recover the gap. Together with Section 5 (Table 3) this localizes the value of any correction-style approach: it can remove pixels, but it cannot supply the coefficient directions the native head fails to realize.

7. Limitations

Single checkpoint and single architecture family; COCO val2017 images were used repeatedly during development, so no result is a blind test; oracle quantities are GT-assisted diagnostics and not method performance; candidate–GT associations are diagnostic, not COCO matching; no accuracy is claimed. The recoverability measurement’s exact objective and the projection audit’s margins are frozen in the project record and cited accordingly. Cross-model generality is untested.

8. Conclusion

Mask-quality failures in a prototype-based segmenter are largely recoverable inside its own frozen representation, and the native prediction does not realize them. The exclusion chain in this paper moves the question from “is there information” to “how must coefficients be realized”, and gives the next method attempt a specific target and a set of controlled comparisons it must beat.

Data and code availability

All readouts use the frozen official YOLO26m-Seg checkpoint and the public COCO validation split with original annotations; no weights are trained in this study. Every number in Tables 1–3 traces to a retained run record in the project’s experiment archive. Per experiment we retain the protocol frozen before execution, per-candidate and per-GT records, independent verification runs with their checks, source snapshots with hashes, and transfer manifests; a machine-readable claim-to-record map (evidence map) enumerates each claim, its source records, and its verification status. The recoverability re-derivation and its verification are retained as separate runs of the same archive.

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Table 2. Recoverability of strict argmax-conditioned mask failures inside the frozen prototype space: an oracle coefficient correction (Eq. 1) reaches Mask75. For each stratum: the definition-complete re-derivation of this study (primary) and the original panel’s recorded values (partially recorded decode conventions; both reported per the frozen protocol). Three degenerate GTs are excluded from the re-derived population.

stratum	re-derived (this study)		recorded (original panel)	
	recov.	share	recov.	share
small (<32 ²)	2,817	72.5%	2,285	58.8%
medium (32 ² –96 ²)	1,266	72.7%	1,237	71.0%
large (≥96 ²)	760	91.7%	741	89.4%
all	4,843	75.0%	4,263	66.0%

Table 3. Exclusion chain: each row removes one candidate explanation of why the native model does not realize the recoverable coefficient directions. Numbers are image-macro or per-instance readouts with bootstrap intervals reported in the project record; scope limits are part of each result.

layer	verdict	key number	scope limit
position selection	excluded	oracle best–median 1.293 points (1.854 on multi-row instances)	diagnostic, fixed panel
supervision amount	excluded	teacher vs same-budget control +0.013 pp, target-group CI crosses 0	253 dev images, 1,816 candidates
ownership readout	excluded	true-ROI vs equal-capacity control: IoU -0.001 , CI 0	8×8 soft ownership, 196 images
ownership via solver	excluded	external evidence through fixed solver -4.99 pp macro IoU	fixed solver parameters, one external model
shared final layer	partial	46.8% of diagnostic gap; +0.055 image-macro IoU, CI positive	mask-BCE diagnostic on frozen features
global correction	bounded	+0.092 pp, CI [+0.022, +0.167]	504 images, 3,560 fixed positives
oracle distillation	excluded	held-out alignment -0.03 (P5 -0.12); repair gain equals control; damage 6.7–7.3%	1 seed, 3 epochs, 796 fit / 197 dev images
constrained projection	excluded	retention 30.2% vs random support 58.8% (gate 80%)	fixed margins and penalty grid